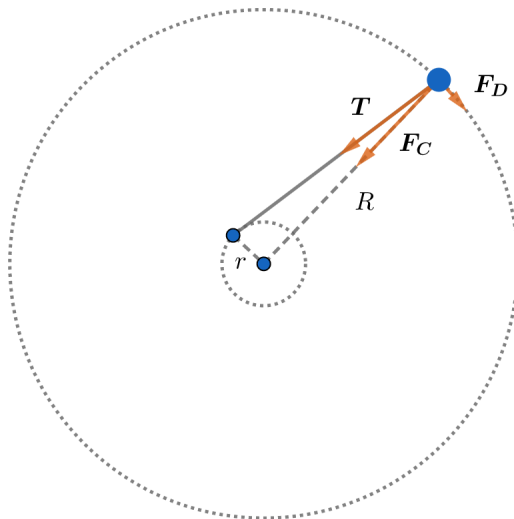


2019B F=ma Exam: Problem 15

Kevin S. Huang



We have tension and air resistance providing the centripetal force. From similar triangles,

$$\frac{F_D}{F_C} = \frac{r}{R}$$
$$F_D = \frac{r}{R} M \omega^2 R = M \omega^2 r = 0.3 \text{ N}$$

so the answer is D.